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Numerical Study of Non-Fourier Heat Flux Across Hybrid Nanofluid Flow Over a Sphere When Thermal Radiation and Chemical Reaction are Substantial

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Abstract. Dynamics of thermal radiation and the Lorentz force on the free convective flow of hybrid nanofluid ($H_2O + Fe_3O_4 + MWCNT$) flow by a solid sphere with chemical reaction are examined in this paper. Non-Fourier heat flux (Cattaneo-Christov heat flux) model is included in the energy equation. Matlab's built-in function, bvp4c, is used to solve governing equations after they have been transformed into ordinary differential equations. Graphs are used to display results (such as changes in the temperature profile caused by thermal radiation). Major findings are, as the volume fraction of nanoparticles increases, the shear stress decreases by 0.16509, while the shear stress increases by 4.593595 with respect to magnetic field. It is noticed that the fluid temperature rises when thermal radiation upsurges. Temperatures drop as a function of thermal relaxation parameter, which is also observed. The data also shows that as Schmidt number rises, so does the mass transfer rate, with an increment rate of 3.356359.

Keywords: Sphere, Hybrid nanofluid, Thermal Radiation, Chemical Reaction, Cattaneo-Christov heat flux, bvp4c.

INTRODUCTION

Initially, Choi [1] coined the term nanofluid. Later, Xuan and Li [2] demonstrate that the thermal conductivity of nanofluids rises dramatically as the volume fraction of small particles ameliorates by taking Water and Copper nanofluid. They mentioned that, when the volume fraction of small particles is amplified from 2.5 to 7.5 percent, the nanofluid's thermal conductivity to that of the base liquid upsurges from 1.24 to 1.78. Li and Kleinstreuer [3] observed that nanofluids containing extremely small volumes of metal nanoparticles are good coolants and resulted in an additional entropy production reduction at lesser Reynolds numbers due to their superior thermal characteristics. Numerous writers [4-6] recently explored diverse nanofluid flows in a variety of geometries, including a revolving cone. Firstly, Suresh et al. [7] created a hybrid nanofluid with a 0.1 percent volume concentration of nanomaterials (Al_2O_3, Cu) in pure water and investigated its heat transfer properties in a straight circular tube. They discovered that suspension of Al_2O_3, Cu nanoparticles enhance the heat transfer performance when compared to pure water. The hybrid nanofluid flow in a straight circular tube was simulated using two distinct approaches by Moghadassi et al. [8]. When nanoparticles are added, the coefficient of friction is increased is one of their findings. Hybrid nanofluid flow ($Cu - Al_2O_3/H_2O$) across an elongating sheet was studied by Devi and Devi [9]. They discovered that Cu/H_2O had a lower heat transfer rate than $Cu - Al_2O_3/H_2O$. Rostami et al. [10] used a flat plate to create dual solutions to the hybrid nanofluid ($Al_2O_3 - SiO_2/H_2O$) stagnation point flow and discovered that the first solution had smaller velocity than the second solution. According to Jarray et al. [11], the higher the permeability, the faster the hybrid nanofluid ($Ag - MgO/H_2O$) flow may transfer heat across the permeable horizontal channel. In a hybrid

nanofluid flow amongst parallel plates with viscous dissipation, Muhammad et al. [12] performed an irreversibility study using bvp4c. They found that when the Eckert number increased, the entropy production increased as well. Several writers [13-18] have recently examined different hybrid nanofluid flows across a wide range of geometries.

Using a Cattaneo-Christov heat flux model with a magnetic field parameter, no one has yet studied the effects of thermal radiation and chemical reaction on hybrid nanofluid flow ($H_2O + Fe_3O_4 + MWCNT$). We intend to close this knowledge gap by utilizing a water-based hybrid nanofluid with thermal radiation and Lorentz forces. When trying to understand how mass transportation works, it is important to consider chemical reactions. The effects of various factors on the flow are depicted in graph form [19-20].

FORMULATION

In this work, laminar flow of a hybrid nanofluid ($H_2O + Fe_3O_4 + MWCNT$) by a sphere with thermal radiation and chemical reaction is analyzed. Energy equation is extended using the heat flow model of Cattaneo-Christov. To begin, let's assume that the x – coordinate is then measured along that surface, y – coordinate is the distance measured perpendicular to that surface. Normal to the surface, a magnetic field with a strength of B_0 is applied (see Fig. 1). Nanomaterials' thermo-physical characteristics are shown in Table 1. Fluid flow is only taken into account approaching the point of stagnation. We also anticipated the induced magnetic field to be discarded.

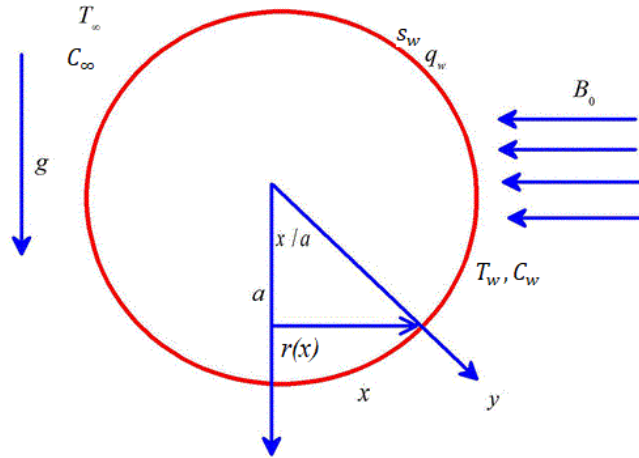


Figure 1. Flow diagram

The following equations govern this problem based on these assumptions: [21-24]

$$\frac{\partial(ru)}{\partial x} + \frac{\partial(rv)}{\partial y} = 0 \quad (1)$$

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = \frac{\mu_{hnf}}{\rho_{hnf}} \frac{\partial^2 u}{\partial y^2} + g \beta_T (T - T_\infty) \sin\left(\frac{x}{a}\right) - \frac{\sigma_{hnf} B_0^2}{\rho_{hnf}} u + g \beta_C (C - C_\infty) \sin\left(\frac{x}{a}\right) \quad (2)$$

$$u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} + \delta^* \left[u^2 \frac{\partial^2 T}{\partial x^2} + v^2 \frac{\partial^2 T}{\partial y^2} + 2uv \frac{\partial^2 T}{\partial x \partial y} + \left[u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \right] \frac{\partial T}{\partial x} + \left[u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} \right] \frac{\partial T}{\partial y} \right] \\ = \frac{k_{hnf}}{(\rho C_p)_{hnf}} \frac{\partial^2 T}{\partial y^2} + \frac{16\sigma^* T_\infty^3}{3k^*} \frac{1}{(\rho C_p)_{hnf}} \frac{\partial^2 T}{\partial y^2} \quad (3)$$

$$u \frac{\partial C}{\partial x} + v \frac{\partial C}{\partial y} = D_m \frac{\partial^2 C}{\partial y^2} - k_1 (C - C_\infty)^n \quad (4)$$

with the conditions

$$\left. \begin{array}{l} \text{at } y = 0 : v = u = 0, T = T_w, C = C_w \\ \text{as } y \rightarrow \infty : u \rightarrow 0, T \rightarrow T_\infty, C \rightarrow C_\infty \end{array} \right\} \quad (5)$$

Note that the radial distance between the harmonic axis and the sphere's surface corresponds to the radius of the sphere a is $r(x) = a \sin\left(\frac{x}{a}\right)$.

THERMO PHYSICAL CHARACTERISTICS OF HYBRID NANOFLUID

$$\left. \begin{array}{l} (\rho C_p)_{hnf} = \left[(1-\phi_1)(\rho C_p)_f + \phi_1(\rho C_p)_{s_1} \right] (1-\phi_2) + (\rho C_p)_{s_2} \phi_2, \\ \mu_{hnf} = \frac{\mu_f}{(1-\phi_1)^{2.5} (1-\phi_2)^{2.5}}, \rho_{hnf} = \left[(1-\phi_1)\rho_f + \phi_1\rho_{s_1} \right] (1-\phi_2) + \rho_{s_2}\phi_2, \\ k_{hnf} = k_{nf} \times \frac{k_{s_2} + 2k_{nf} - 2k_{nf}\phi_2 + 2k_{s_2}\phi_2}{k_{s_2} + 2k_{nf} + k_{nf}\phi_2 - k_{s_2}\phi_2}, k_{nf} = k_f \times \frac{k_{s_1} + 2k_f - 2k_f\phi_1 + 2k_{s_1}\phi_1}{k_{s_1} + 2k_f + k_f\phi_1 - k_{s_1}\phi_1}, \\ \sigma_{hnf} = \sigma_{nf} \times \frac{\sigma_{s_2} + 2\sigma_{nf} - 2\sigma_{nf}\phi_2 + 2\sigma_{s_2}\phi_2}{\sigma_{s_2} + 2\sigma_{nf} + \sigma_{nf}\phi_2 - \sigma_{s_2}\phi_2}, \sigma_{nf} = \sigma_f \times \frac{\sigma_{s_1} + 2\sigma_f - 2\sigma_f\phi_1 + 2\sigma_{s_1}\phi_1}{\sigma_{s_1} + 2\sigma_f + \sigma_f\phi_1 - \sigma_{s_1}\phi_1} \end{array} \right\}.$$

TABLE 1. Values of density and other properties of nanomaterials and base fluid (Mehryan et al. [22])			
S. No		$Fe_3O_4 (\phi_1)$	$MWCNT (\phi_2)$
1	$\rho (kgm^{-3})$	5810	2100
2	$C_p (J(kg K)^{-1})$	670	711
3	$k (W(m K)^{-1})$	6	3000
4	$\sigma (Sm^{-1})$	2.5×10^4	1.9×10^{-4}

The controlling equations are renovated as a series of ODEs using the similarity transformations shown below (Amanulla et al., [19]):

$$\left. \begin{array}{l} \eta = \frac{y}{a} (Gr)^{1/4}, \psi = \nu \xi (Gr)^{1/4} f(\xi, \eta), \xi = \frac{x}{a}, \\ T = T_\infty + (T_w - T_\infty) \theta(\xi, \eta), \Phi(\xi, \eta) = \frac{C - C_\infty}{C_w - C_\infty} \end{array} \right\} \quad (6)$$

It is believed that u, v are velocity components that are given by

$$u = \frac{1}{r} \frac{\partial \psi}{\partial y}, v = -\frac{1}{r} \frac{\partial \psi}{\partial x}, \quad (7)$$

so that the continuity equation (1) can be satisfied.

Equations (2-4) are transmuted as follows using (6) and (7):

$$\frac{1}{A_1 A_2} f''' - \left(\xi \left(f' \frac{\partial f'}{\partial \xi} - f'' \frac{\partial f}{\partial \xi} \right) - ff'' (1 + \xi \cot \xi) + f'^2 \right) + \frac{\sin \xi}{\xi} \theta - \frac{A_3 M}{A_2} f' + \lambda \frac{\sin \xi}{\xi} \Phi = 0 \quad (8)$$

$$\left(\frac{A_4}{A_5 \text{Pr}} + \frac{Ra}{A_4 \text{Pr}} \right) \theta'' - \left(\xi \left(f' \frac{\partial \theta}{\partial \xi} - \theta' \frac{\partial f}{\partial \xi} \right) - f (1 + \xi \cot \xi) \theta' \right) = \Gamma \left[\begin{aligned} & \xi^2 f'^2 \frac{\partial^2 \theta}{\partial \xi^2} + f^2 \theta'' + \xi^2 \left(\frac{\partial f}{\partial \xi} \right)^2 \theta'' + 2 \xi f \frac{\partial f}{\partial \xi} \theta'' - 2 \xi ff' \frac{\partial^2 \theta}{\partial \xi \partial \eta} + 2 \xi^2 f' \frac{\partial f}{\partial \xi} \frac{\partial^2 \theta}{\partial \xi \partial \eta} \\ & + \xi^2 f' \frac{\partial^2 f}{\partial \xi \partial \eta} \frac{\partial \theta}{\partial \xi} + \xi f'^2 \frac{\partial \theta}{\partial \xi} - \xi^2 f'^2 \cot \xi \frac{\partial \theta}{\partial \xi} - \xi ff'' \frac{\partial \theta}{\partial \xi} - \xi^2 \frac{\partial f}{\partial \xi} \frac{\partial \theta}{\partial \xi} f'' \\ & - 2 \xi f' \theta' \frac{\partial f}{\partial \xi} - \xi^2 f' \theta' \frac{\partial^2 f}{\partial \xi^2} + \xi \cot \xi ff' \theta' + \xi^2 \cot \xi f' \theta' \frac{\partial f}{\partial \xi} + ff' \theta' \\ & - \xi^2 f' \theta' \frac{\partial f}{\partial \xi} \cot \xi + ff' \theta' + \xi f \theta' \frac{\partial^2 f}{\partial \eta \partial \xi} + \xi f' \theta' \frac{\partial f}{\partial \xi} + \xi^2 \frac{\partial f}{\partial \xi} \frac{\partial^2 f}{\partial \eta \partial \xi} \theta' \end{aligned} \right] \quad (9)$$

$$\xi \left(f' \frac{\partial \Phi}{\partial \xi} - \Phi' \frac{\partial f}{\partial \xi} \right) - f (1 + \xi \cot \xi) \Phi' = \frac{1}{St} \Phi'' - Cr \Phi^n \quad (10)$$

and '(prime) denotes the derivative with respect to η .

Because the flow is close to the sphere's stagnation point, $\xi \approx 0$.

Equations (8-10) may be rewritten with this condition as:

$$\frac{1}{A_1 A_2} f''' + \theta - \frac{A_3 M}{A_2} f' + \lambda \Phi + 2 ff'' = 0 \quad (11)$$

$$\left(\frac{A_4}{A_5 \text{Pr}} + \frac{Ra}{A_5 \text{Pr}} \right) \theta'' + 2 f \theta' - \Gamma \left[f^2 \theta'' + 2 ff' \theta' \right] = 0 \quad (12)$$

$$\frac{1}{St} \Phi'' + 2 f \Phi' - Cr \Phi^n = 0 \quad (13)$$

and conditions in (5) can be rewritten as

$$\left. \begin{aligned} \text{at } \eta = 0: & f = 0, f' = 0, \theta = 1, \Phi = 1, \\ \text{as } \eta \rightarrow \infty: & f' \rightarrow 0, \theta \rightarrow 0, \Phi \rightarrow 0 \end{aligned} \right\}, \quad (14)$$

where

$$\left. \begin{aligned}
A_1 &= (1-\phi_2) \left[(1-\phi_1) + \phi_1 \frac{\rho_1}{\rho_f} \right] + \phi_2 \frac{\rho_2}{\rho_f}, A_2 = (1-\phi_1)^{2.5} (1-\phi_2)^{2.5}, \\
A_{31} &= \frac{\sigma_1 + 2\sigma_f - 2\phi_1(\sigma_f - \sigma_1)}{\sigma_1 + 2\sigma_f + \phi_1(\sigma_f - \sigma_1)}, \\
A_3 &= \frac{\sigma_2 + 2A_{31}\sigma_f - 2\phi_2(A_{31}\sigma_f - \sigma_2)}{\sigma_2 + 2A_{31}\sigma_f + \phi_2(A_{31}\sigma_f - \sigma_2)} A_{31}, A_{41} = \frac{k_1 + 2k_f - 2\phi_1(k_f - k_1)}{k_1 + 2k_f + \phi_1(k_f - k_1)}, \\
A_4 &= \frac{k_2 + 2A_{41}k_f - 2\phi_2(A_{41}k_f - k_2)}{k_2 + 2A_{41}k_f + \phi_2(A_{41}k_f - k_2)} A_{41}, A_5 = (1-\phi_2) \left[(1-\phi_1) + \phi_1 \frac{(\rho C_p)_1}{(\rho C_p)_f} \right] + \phi_2 \frac{(\rho C_p)_2}{(\rho C_p)_f} \\
M &= \frac{a^2 \sigma_f B_0^2}{\rho \nu Gr^{0.5}}, Pr = \frac{\mu C_p}{k_f}, \lambda = \frac{Gc}{Gr}, Gc = \frac{g \beta_c (C_w - C_\infty) a^3}{\nu^2}, \\
Gr &= \frac{g \beta_r (T_w - T_\infty) a^3}{\nu^2}, St = \frac{\nu}{D_m}, Ra = \frac{16 \sigma^* T_\infty^3}{3k^* k}, \\
Cr &= \frac{(C_w - C_\infty)^{n-1} k_1 a^2}{\nu Gr^{0.5}}, \Gamma = \frac{\delta^* \nu}{r a^2} Gr^{0.5}
\end{aligned} \right\} \text{ and }$$

Engineering parameters of interest

Shear stress is defined as:

$$Cf = \frac{Gr^{-\frac{3}{4}} a^2}{\mu \nu} \tau_w \quad (15)$$

$$\text{where } \tau_w = \mu_{hnf} \left(\frac{\partial u}{\partial y} \right) \Big|_{y=0}$$

By using (6) & (7), term in (15) can be rewritten as

$$Cf = \frac{1}{A_1} \xi f''(\xi, 0).$$

Formulae to find transfer rates (heat and mass) are chosen as:

$$Nu = \frac{a Gr^{-\frac{1}{4}}}{k_f (T_w - T_\infty)} q_w, Sh = \frac{a Gr^{-\frac{1}{4}}}{D_m (C_w - C_\infty)} s_w \quad (16)$$

$$\text{where (wall heat flux) } q_w = - \left(k_{hnf} + \frac{16 \sigma^* T_\infty^3}{3k^*} \right) \frac{\partial T}{\partial y} \Big|_{y=0}, \text{ (wall mass flux) } s_w = - D_m \frac{\partial C}{\partial y} \Big|_{y=0}.$$

with the aid of (7), formulae in (16) are rewritten as:

$$Nu = -A_4 (1 + Ra) \theta'(\xi, 0) \text{ and } Sh = -\Phi'(\xi, 0).$$

DISCUSSION OF RESULTS

Bvp4c was used to solve the equations (11-13) and the conditions (14) in conjunction with the conditions (14). The results are presented in the form of graphs.

Multiple Linear Regression. It is possible to determine the correlation between at least two independent and one dependent variable using multiple linear regression. For y denotes the dependent variable, x_i represent the independent variable, a_0 is y -intercept, and a_i is the Regression coefficient of x_i for $i = 1, 2, \dots, n$. Multiple linear regression formula is as follows:

$$y = a_0 + a_1x_1 + a_2x_2 + \dots + a_nx_n.$$

While getting the values of a_i 's, if p -value is less than 0.05 then the relation between y and x_i is significant. The following models were employed to determine the link between engineering factors of interest, such as shear stress, and parameters such as magnetic field parameter.

$$Cf = A_0 + A_2M + A_1\phi_1 \quad (17)$$

$$Nu = B_0 + B_1M + B_2Ra \quad (18)$$

$$Sh = C_0 + C_1St + C_2Cr \quad (19)$$

We were able to get the following findings using 30 sets of values for each equation:

$$Cf = 1.0364 - 0.1858M + 4.61\phi_1 \quad (20)$$

$$Nu = 0.6765 - 0.0644M + 0.3314Ra \quad (21)$$

$$Sh = -0.6489 + 3.3569St + 0.1078Cr \quad (22)$$

The volume fraction of Fe_3O_4 and the skin friction coefficient are shown to have a positive association (see Equation (20)). Furthermore, an increase in the magnetic field parameter is associated with a decrease in shear stress. In the first situation, the rate of increment is 4.593595, while the rate of decrease is -0.16509 in second case. There is an increase in heat transmission with increased thermal radiation, and a decrease in the rate of heat transfer with an increase in the magnetic field parameter. In the first situation, the increment is 0.319268, whereas the decrement is -0.05803 in second case. Mass transfer rate increases with an increase in Schmidt number and chemical reaction parameter (see equation 22). The increment rates are 3.356359 and 0.258335, according to the data. **Regular profiles including Velocity.** Applying a magnetic field can change the way fluid moves. Particles in the liquid form a chain that moves in the direction of an attractive field. They bump into each other during this time, which makes it hard for the fluid to flow. As a result, the speed of the fluid slows down and temperature of the rises because the viscosity of the fluid goes up (Figs. 2 & 3). As the volume fraction of nanoparticles grows, more fluid particles come into contact with each other. As a result, the temperature of the fluid goes up (Fig. 4). Depending on the type of fluid, some or all of the heat from the sun is absorbed by the fluid, increasing the energy of the particles in the fluid. The particles hit each other, which causes the temperature of the fluid to rise (Fig. 5). Thermal relaxation is a term used to describe how long it takes for fluid temperatures to drop at least half of their maximum value. So, the higher the thermal relaxation parameter, the lower the temperature of the fluid will be, because it takes longer for the temperature of the fluid to drop (Fig. 6). When the chemical reaction parameter is increased, concentration often decreases (Fig. 7). Entropy is increasing, which could be the source of this. When St rises, the fluid's momentum diffusivity exceeds its mass diffusivity by a large margin, indicating that it is moving quickly and its concentration is dropping (Fig. 8).

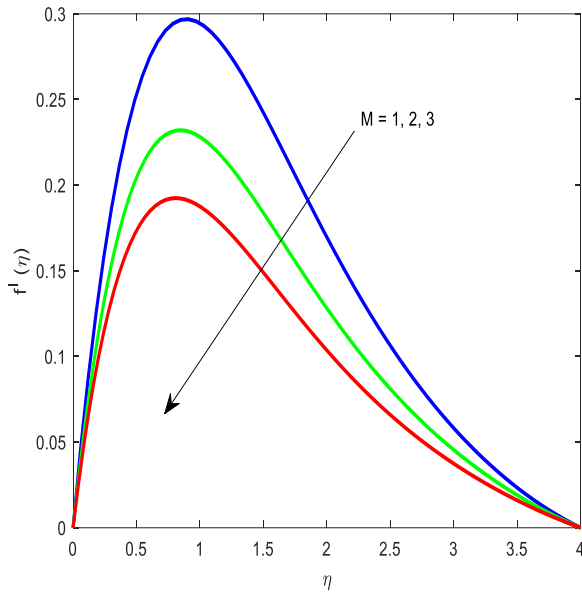


FIGURE 2.Effects on velocity profile due to M

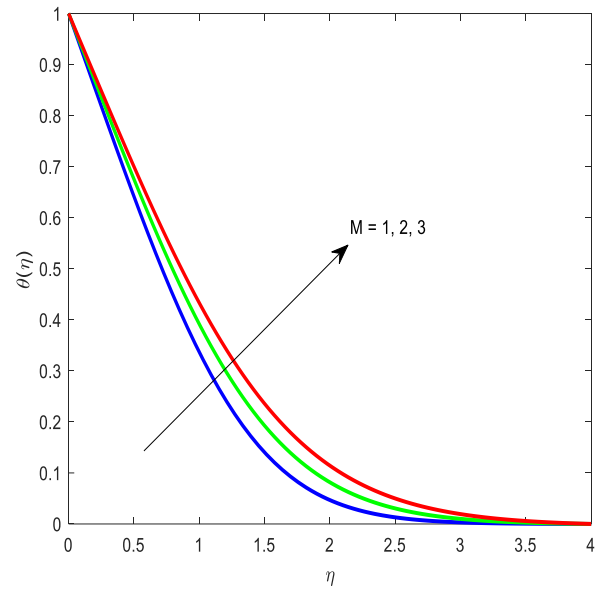


FIGURE 3. Effects on temperature profile due to M

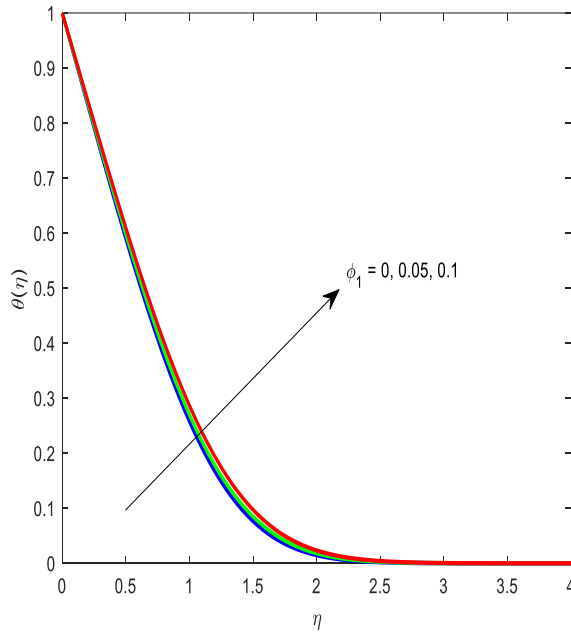


FIGURE 4.Effects on temperature profile due to ϕ_1

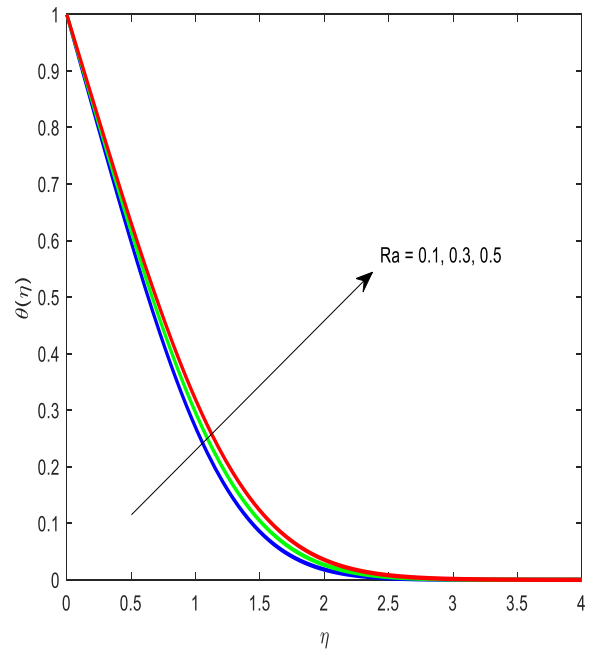


FIGURE 5.Effects on temperature profile due to Ra

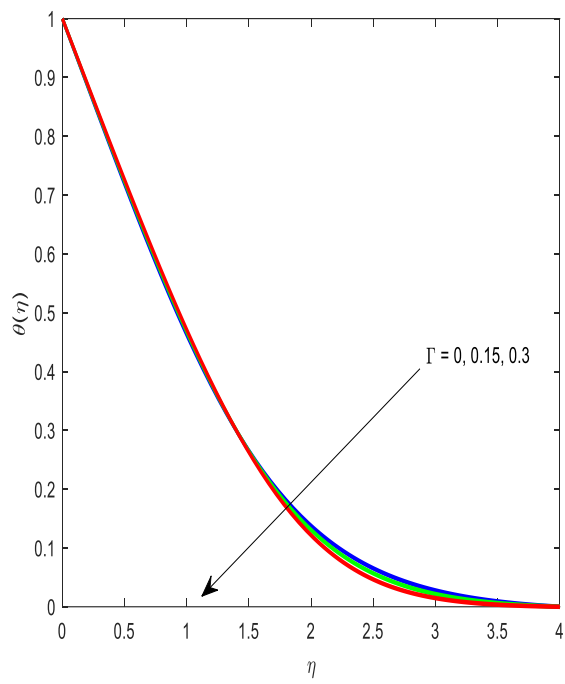


FIGURE 6. Effects on velocity profile due to Γ

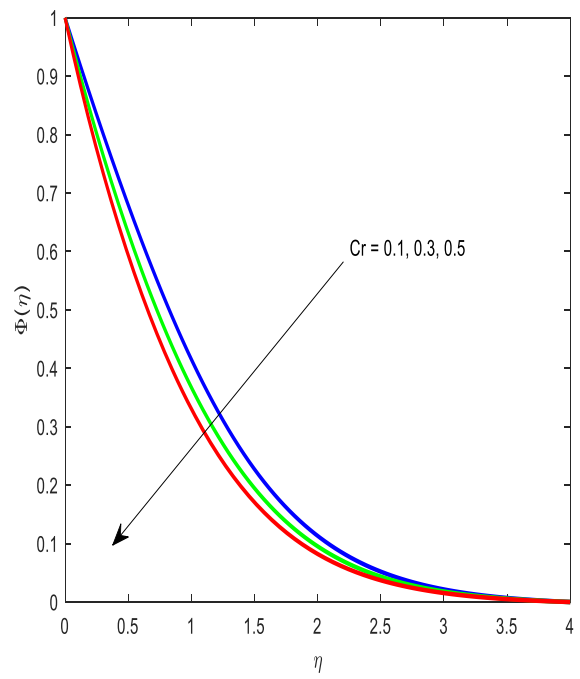


FIGURE 7. Effects on concentration profile due to Cr

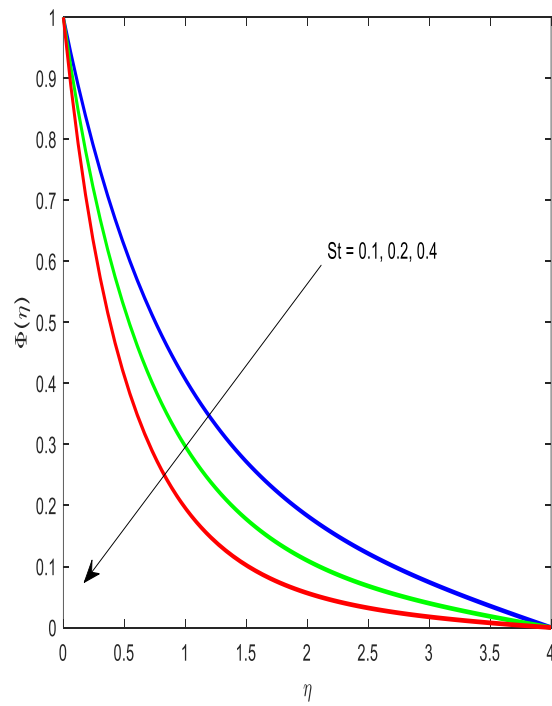


FIGURE 8. Effects on concentration profile due to St

CONCLUSIONS

The effects of chemical reaction and Lorentz forces on the flow of $Fe_3O_4 + MWCNT/H_2O$ across the surface of a sphere exposed to the thermal radiation with Cattaneo-Christov heat flux are depicted graphically in this article. The following are some of the study's key findings:

- The volume fraction of Fe_3O_4 and the skin friction coefficient are shown to have a positive association. In this situation, the rate of increment is 4.593595.
- An increase in the magnetic field parameter is associated with a decrease in shear stress, where the rate of decrease is -0.16509.
- Rise in thermal radiation causes the temperature of the fluid to rise.
- When the chemical reaction parameter is increased, concentration often decreases.
- Mass transfer rate increases with an increase in Schmidt number. The increment rate is 3.356359 according to the data.

NOMENCLATURE

μ Dynamic viscosity of the fluid $[kg\ m^{-1}s^{-1}]$	Φ Dimensionless concentration of fluid
St Schmidt number	Gr Local thermal Grashoff number
ρ Density of the fluid $[kgm^{-3}]$	Gc Local diffusion Grashoff number
Pr Prandtl number	λ Buoyancy ratio parameter
β_T Thermal expansion coefficient	D_m Molecular diffusivity $[m^2s^{-1}]$
k^* Mean absorption coefficient	M Magnetic field Parameter
g Acceleration of gravity	ξ Dimensionless coordinate
ν Kinematic viscosity $[m^2s^{-1}]$	β_C Diffusion expansion coefficient
T Dimensional temperature of fluid $[K]$	θ Non-dimensional temperature
k Thermal conductivity $[Wm^{-1}K^{-1}]$	k_1 Chemical reaction rate
C Dimensional concentration of fluid $[mol\ m^{-3}]$	Cr Non dimensional chemical reaction parameter
f' Dimensionless velocity	σ Electrical conductivity
C_p Heat capacity	ρ Density
ψ Stream function	n Order of chemical reaction
u, v Velocity components in x, y directions $[ms^{-1}]$	Subscripts
δ^* Thermal relaxation time of heat flux	f Fluid
Γ Non dimensional thermal relaxation time parameter	nf Nanofluid
η Similarity variable	hnf Hybrid nanofluid
σ^* Stefan-Boltzmann constant	
f Dimensionless Stream function	

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